

----- Active space -----

ABNORMALITIES OF STATURE

khanirfan0392@gmail.com

Short stature

00:00:18

Height for age $< 3rd$ percentile or $< -2SD$.

TYPES :

	Normal variant of short stature	Pathological short stature
Height for age	< -2 to -3 SD	< -3 SD
Seen in	• Constitutional delay (Overall m/c cause). • Familial short stature.	Syndromic presentations
Prepubertal growth velocity	4-6 cm/yr (Normal)	4-6 cm/yr

ETIOLOGY :

1. Chronic undernutrition : **Stunting**.
2. Chronic systemic illness : Hampers growth.
Eg : Chronic heart or lung diseases.
3. Endocrine disorders :
 - a. Hormonal deficiency :
 - Growth hormone deficiency.
 - Thyroid hormone deficiency.
 - b. Hormone excess :
 - Cushing syndrome.
 - Precocious puberty : Excess of sex steroids.
4. Bony disorders : Bone growth affected.
Eg : Achondroplasia : Short proximal part of limbs
↓
Rhizomelia
5. Genetic disorders :
 - Down's syndrome.
 - Turner's syndrome.

↓
Premature fusion of epiphysis
(Arrests bone growth).
↓
Short stature.

STEPS OF EVALUATION :

1. Accurate assessment of height :
use stadiometer/infantometer.
(> 2 yrs old) (< 2 yrs old)
2. Compare with growth charts :
Plot values and assess percentiles.

3. mid parental height (MPH) :

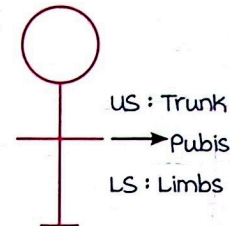
----- Active space -----

- Target height of the child.
- Gives an approximate of the child's adult height.

$$\text{MPH} = \left[\frac{(\text{Mother's} + \text{Father's}) \text{ height}}{2} \right] \begin{cases} + 6.5 \text{ cm (boys)} \\ - 6.5 \text{ cm (girls)} \end{cases}$$

4. Assessment of body proportion :

- By upper segment : Lower segment ratio (US : LS).
- upper segment : Height of the trunk.
- Lower segment : Height/length below pubis.



Normal value :

At birth : $17:1$ → 3 years : $1.3:1$ → At 10 yrs : $1:1$ → Adults : $0.9:1$

Increasing length of lower segment

Abnormalities in US : LS ratio :

1. ↑ US : LS

Lower segment/limbs are short.

- Achondroplasia.
- Rickets.
- Congenital hypothyroidism.

2. ↓ US : LS

- Trunk is smaller → Pathology of vertebral column.
- Eg : Spondyloepiphyseal dysplasia.

5. Bone age : Evaluated by X-Ray.

Depends on appearance of ossification centres.

Age range	X-Ray
Infancy	Shoulder X-Ray
1-13 yrs	Left wrist (Ossification centres of carpal bones)

1. Age assigned after comparison with atlas

- Tanner Whitehouse method.
- Greulich & Pyle atlas (G.P.).

2. Bone expert software.

Interpretation :

- In most cases of short stature → Bone age (BA) < Chronological age (CA).
- **Exception** : BA = CA : Familial short stature.
- BA > CA : Precocious puberty (D/t ↑ sex steroids).

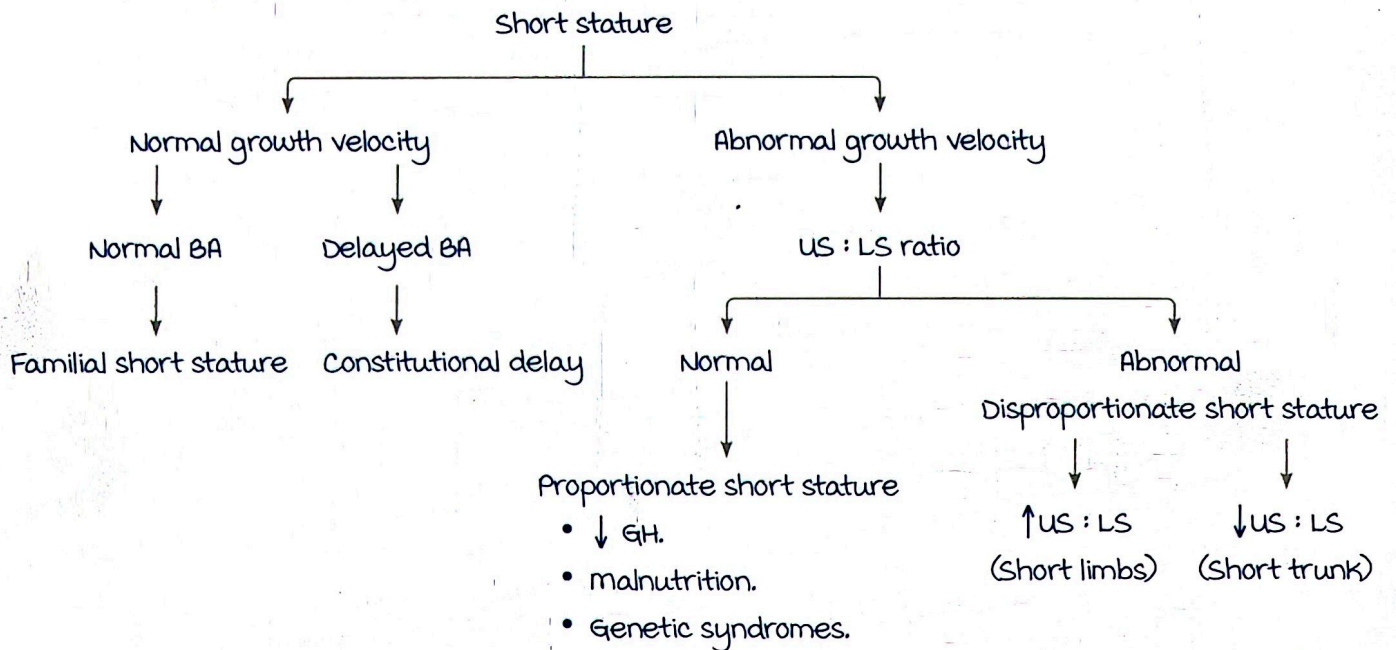
khanirfan0392@gmail.com

NORMAL VARIANT OF SHORT STATURE :

D/t genetic potential.

	Constitutional delay	Familial short stature
Final (adult) height	Normal	Short
Age at puberty	Delayed	Normal
Bone age	Delayed	Normal
Parents' height	Normal	Short

----- Active space -----

ALGORITHMIC APPROACH TO SHORT STATURE :**Tall stature**

00:22:35

648c85cfee3b03a74e182fab

Height for age $> +2S.D$ or 97th percentile.**NORMAL VARIANT :**

- m/c type.
- Types
 - Familial
 - Constitutional acceleration of growth (CAG)

PATHOLOGICAL CAUSES :**Syndromes :**

- marfan syndrome : Superior lens dislocation.
- Klinefelter's syndrome (Genetic).
- Overgrowth/cerebral gigantism : Sotos syndrome.

metabolic disorder :

- Homocystinuria : D/t cystathionine $-\beta$ synthase deficiency.
 - Inferior lens dislocation.
 - marfan like features.

Endocrine :

- Pituitary adenoma : GH excess.
- Hyperthyroidism.